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Evaluation of Modern Hopfield Networks as Associative Memory for Compliance Processes: A Comparative Study with Alternative Retrieval Architectures

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Abstract

This Master Thesis describes the implementation of a solution to control and monitor electric consumers via the Internet. Needs of this solution are defined by use of the Kano model. Existing technologies and standards are benchmarked by means of a cost-utility analysis. The solution that scores the highest value of benefit will be implemented typically.

A Zigbit Module forms the basis of the hardware. It bundles an AVR microcontroller and an IEEE 802.15.4 transceiver. Together with a self-dimensioned capacitive power supply it is mounted in a socket housing.

To be future-proof, the IPv6 protocol is used. The 6LoWPAN protocol handles the adaptation layer. Contiki is used as operating system. It is delivered with a ready-to-use web server application which is customized for the own purposes. The IEC 60870-5-104 protocol is implemented from scratch. It is based on TCP/IP and is used in the field of energy management systems. It is particularly suitable for automated access.

Via a public address given by IPv6 tunnel broker SixXS the solution is accessible worldwide. The electric consumer can be switched on and off by the means of a web browser or an energy management system.

The needs according to the Kano model are almost completeley achieved. It is possible to implement a solution consisting of a web server and IEC 60870-5-104 application in ressource constraint environments. Possible applications for such a solution are in the field of home automation and smart grid.

Chapter 1: Introduction

1.1 Research Background

The rapid growth of the RegTech industry is driven by the increasing complexity, volume, and frequency of regulatory requirements. Regulatory information is often contained in large collections of semi-structured and semi-labeled documents, where relevant knowledge is distributed across text, metadata, references, and hierarchical structures. Traditional compliance systems face challenges in efficiently retrieving and connecting this heterogeneous information.

A fundamental challenge is the interconnected nature of regulatory knowledge. Regulations frequently reference other regulations, standards, technical specifications, and exceptions, forming complex networks of dependencies. As a result, understanding a specific requirement often requires identifying relationships between entities and retrieving information across multiple sources. Conventional keyword-based and document-level retrieval methods are insufficient for capturing these semantic connections.

Recent advances in artificial intelligence, including Transformer-based models, Retrieval-Augmented Generation (RAG), and associative memory architectures, provide new opportunities for representing and retrieving interconnected regulatory knowledge. This thesis investigates approaches for improving regulatory information retrieval through semantic representation and associative retrieval methods, with a focus on entity-aware knowledge retrieval in RegTech applications.

1.2 Problem Statement

Regulatory knowledge is inherently interconnected, with legal documents frequently referencing other regulations, directives, standards, and technical specifications. These references form complex networks of dependencies in which understanding a specific legal requirement often requires traversing relationships between multiple documents. Previous work has highlighted that legal knowledge can be represented as interconnected networks or graphs, where documents and their references form nodes and edges that capture regulatory relationships.

A major challenge in establishing aforementioned graphs, is that the same regulation may be referenced using different surface forms throughout different legal texts. References may use the official title, legal identifier, common abbreviations, or descriptive names. For example, the regulation formally known as *Regulation (EU) 2016/679* may be referenced as *Regulation (EU) 2016/679*, *General Data Protection Regulation*, or *GDPR* depending on the context. Similarly, references to amending acts introduce additional ambiguity, as legal documents may refer to both the original regulation and subsequent amendments. For example, a legal text may reference *Regulation (EU) 2016/679 (GDPR)* while another refers to a later regulation that modifies specific provisions of the GDPR. These cases create difficulties for automated retrieval systems, which must distinguish between the original legal entity, amending legislation, consolidated versions, and related documents.

Traditional keyword-based retrieval systems struggle with these challenges because they primarily rely on lexical overlap and often cannot distinguish between closely related legal entities. A search for a regulation identifier or title may retrieve the original regulation, an amending regulation, a consolidated version, or related documents without understanding the relationship between them. As a result, retrieving relevant regulatory knowledge requires methods capable of recognizing semantic equivalence between references and linking them to the correct underlying entities.

Therefore, a key prerequisite for enabling effective traversal and retrieval within regulatory knowledge graphs is reliable entity linking of legal references to specific regulatory entities. This thesis addresses this challenge by investigating AI-based approaches for regulatory entity linking and retrieval, enabling more accurate representation and exploration of interconnected regulatory knowledge.

1.3 Research Gap

Automated compliance workflows rely on the ability to accurately identify and link references within regulatory documents to their corresponding legal entities. However, regulatory entity linking remains a challenging and insufficiently explored domain due to the complex structure of legal texts, variations in regulatory terminology, and the frequent use of multiple references for the same legal act. Existing entity linking approaches are primarily developed and evaluated on general-purpose datasets and do not fully address the characteristics of regulatory knowledge.

Transformer-based models have demonstrated strong capabilities in semantic representation and retrieval, making them a promising approach for regulatory entity linking. However, the mechanisms underlying their associative retrieval capabilities are not yet fully understood from the perspective of regulatory knowledge processing. Recent theoretical work has shown that Transformer attention mechanisms can be interpreted through the lens of Modern Hopfield Networks, where attention updates correspond to associative memory retrieval processes.

Despite this theoretical connection, the implications of associative memory-based interpretations of Transformer architectures for regulatory entity linking and automated compliance workflows remain insufficiently investigated. Understanding how these mechanisms contribute to robust retrieval and decision-making is essential for developing reliable AI-supported compliance systems.

1.4 Research Question

RQ: RQ: To what extent can attention-based retrieval models, motivated by the associative memory framework of Modern Hopfield Networks, improve regulatory entity linking?

To address this question, the following sub-research questions are defined:

RQ1: How can a regulatory reference entity linking framework be designed to address the challenges of regulatory references and semi-structured regulatory data?

RQ2: How can data preparation and augmentation strategies support the development of entity linking models?

RQ3: How effectively can Transformer-based retrieval approaches identify and resolve regulatory references within complex legal documents?

RQ4: How does the proposed framework perform in accurately linking regulatory references to their corresponding entities?

RQ5: How can confidence-based human oversight mechanisms be integrated into the proposed framework to identify cases requiring additional expert review?

1.5 Research Contributions

Following a Design Science Research approach, this thesis contributes a framework for regulatory entity linking that addresses the challenges of identifying and linking references within complex regulatory documents. The developed artifact covers the complete entity linking lifecycle, including regulatory data preparation, data augmentation, representation learning, candidate retrieval, entity resolution, and inference.

The primary contribution of this thesis is the design and evaluation of an end-to-end regulatory entity linking framework capable of handling the characteristics of regulatory knowledge, including heterogeneous document structures, semi-structured and semi-labeled data, limited annotated resources, and varying references to the same legal entities.

The framework further contributes methodological insights into the application of Transformer-based retrieval architectures for regulatory entity linking, including multi-stage retrieval approaches that combine efficient candidate generation with semantic reranking. Additionally, this thesis investigates the role of data augmentation strategies for improving the robustness and scalability of regulatory entity linking in settings where manually annotated legal data is limited.

Finally, this thesis provides a theoretical perspective on the relationship between Transformer architectures and associative memory models, particularly Modern Hopfield Networks, to analyze the associative retrieval mechanisms underlying semantic representations used in regulatory knowledge processing.

1.6 Structure of the Thesis

The remainder of this thesis is structured as follows.

Chapter 2 introduces the theoretical background and discusses relevant concepts. Chapter 3 describes the methodology and implementation of the proposed approach. Chapter 4 presents the evaluation and discusses the results. Finally, Chapter 5 summarizes the findings and provides an outlook for future work.

Chapter 2: Theoretical Framework

2.1 Representation Spaces

A fundamental aspect of machine learning systems is how information is represented internally. Since real-world information such as text, images, and signals is not naturally expressed in a format suitable for mathematical operations, it must be transformed into a structured representation.

A representation space defines how information is encoded and how relationships between individual elements are expressed. Depending on the chosen representation, information can be organized as discrete states or continuous vectors. These approaches differ in how information is stored and how relationships between individual representations can be modeled.

In general, a representation function maps an input sample x from an input space $\mathcal{X}$ into a representation space $\mathcal{Z}$:

$$f : \mathcal{X} \to \mathcal{Z} \tag{2.1}$$

where $\mathcal{Z}$ defines the space in which information is stored and compared.

2.2 Representation Learning

While representation spaces define how information can be encoded and compared, the quality of a representation strongly depends on how well it captures relevant characteristics of the underlying data. Representation learning refers to the process of automatically learning such representations from data rather than relying on manually designed features.

Given a dataset of input samples, a representation learning model learns a mapping function that transforms raw inputs into representations where relevant relationships are preserved and irrelevant variations are minimized. The objective is to learn a representation space in which samples with similar semantic meaning are located closer together, while dissimilar samples are separated.

In the context of natural language processing, representation learning enables the transformation of textual information into dense vector representations, commonly referred to as embeddings. Modern Transformer-based models learn contextualized representations in which the meaning of a token or sequence depends on its surrounding context. These representations can subsequently be adapted through fine-tuning on domain-specific data to better capture task-specific relationships.

For entity linking tasks, representation learning is particularly important because the model must learn semantic relationships between textual mentions and their corresponding entities. Rather than relying on exact lexical overlap, learned representations enable the identification of semantically equivalent references that may use different surface forms.

2.2.1 Discrete Representation Spaces

In discrete representation spaces, information is represented through a finite set of distinct states:

$$\mathcal{Z}_D = \{z_1, z_2, ..., z_n\} \tag{2.2}$$

where each representation z_i corresponds to a specific symbol, category, or pattern. Relationships between different states are not inherently represented, meaning that two states are either identical or distinct.

This type of representation is commonly used in symbolic systems and early associative memory models. For example, classical Hopfield networks represent memories as discrete binary patterns:

$$\xi_i \in \{-1, +1\}^N \tag{2.3}$$

where each stored pattern corresponds to a stable state of the network [Hopfield 1982].

While discrete representations allow for clear separation between individual states, they provide limited capability for expressing gradual relationships or similarities between different representations.

2.2.2 Continuous Representation Spaces

Continuous representation spaces represent information as points within a high-dimensional vector space:

$$\mathcal{Z}_C \subseteq \mathbb{R}^d \tag{2.4}$$

where each representation is encoded as a vector:

$$z = f(x) = [z_1, z_2 \ldots, z_d]^T \tag{2.5}$$

Instead of assigning isolated states to individual concepts or patterns, information is encoded through numerical vectors, where relationships between representations can be expressed through geometric properties such as distance or similarity.

The use of distributed continuous representations has become a central concept in machine learning, where information is represented through patterns of numerical values rather than isolated symbolic states [HS86]. Similar information can be represented by nearby points within the vector space, allowing the representation space itself to capture relationships between stored information.

Within associative memory systems, continuous representation spaces enable memories to be represented as high-dimensional vectors rather than discrete states. A memory set can therefore be expressed as:

$$M = \{m_1, m_2 \ldots, m_k\}, \quad m_i \in \mathbb{R}^d \tag{2.6}$$

where each m_i represents an individual stored memory within the continuous representation space. This provides a more flexible representation of information and forms the basis for modern approaches to associative memory and machine learning systems [Ramsauer et al. 2021].

In this work, continuous representation spaces serve as the foundation for storing and accessing memories.

2.3 Embeddings in Natural Language Processing (NLP)

Natural language is an unstructured form of information that is represented through words, sentences, and context rather than explicit numerical structures. Since machine learning models operate on mathematical representations, NLP systems require a method to transform textual information into a structured format that can be processed computationally. Embeddings provide such a representation by mapping words, sentences, or documents into dense vectors within a continuous vector space, where relationships between textual elements can be represented numerically.

The idea of embeddings is based on the distributional hypothesis, which states that words occurring in similar contexts tend to have related meanings [Har54]. Modern embedding methods use neural networks and large-scale training data to learn vector representations that capture patterns in language usage and contextual relationships.

In contrast to traditional word embeddings, which assign a fixed vector to each word, contextual embedding methods generate representations based on the surrounding text. Models such as BERT use transformer architectures to produce context-dependent representations, allowing the same word to receive different representations depending on its usage [Dev19].

In this work, embeddings are used to transform unstructured textual information into structured numerical representations that can be compared and processed by mathematical models.

2.4 Retrieval in Machine Learning Data Processing Pipelines

A central design question in machine learning systems is where task-relevant knowledge should be stored. In traditional parametric models, knowledge is acquired during training and encoded within the model parameters. Consequently, adapting to new information, updated requirements, or changing domain knowledge generally requires retraining or fine-tuning the model.

Retrieval-based approaches separate learned model behavior from stored knowledge by maintaining an external knowledge source that can be queried during inference. Rather than relying solely on information encoded in the model parameters, the model is provided with additional context retrieved for a given input. This external knowledge can be updated independently of the model, allowing changes to the underlying information source without modifying the trained model itself.

Retrieved information may include, but is not limited to:

- predefined rules,
- domain-specific knowledge,
- previous decisions, or
- precomputed feature embeddings.

Retrieval systems generally consist of two stages: indexing and querying. During indexing, documents, knowledge objects, or data instances are transformed into representations that support efficient search. During inference, an input query is represented in the same embedding space, enabling the retrieval of the most relevant stored information according to a similarity measure.

Traditional retrieval methods rely on lexical matching techniques such as the Vector Space Model and probabilistic ranking algorithms including BM25, which rank documents based on keyword overlap and statistical relevance [Salton and McGill 1983; Robertson and

Zaragoza 2009]. More recent retrieval systems employ dense vector representations, allowing semantically similar queries and documents to be matched even when they do not share the same vocabulary. This enables retrieval based on semantic similarity rather than exact lexical overlap.

In the context of this work, retrieval systems serves as an external memory mechanism that complements the associative memory represented by the model parameters. The distinction between parametric memory and retrieval memory forms the basis for the comparison presented in this thesis.

2.5 Evolution of Hopfield Networks

The concept of associative memory has a long history in artificial neural networks. A foundational contribution was made by John J. Hopfield in 1982 with the introduction of the Hopfield network, a recurrent neural network designed to store and retrieve information through associative memory [Hopfield 1982]. Unlike feed-forward neural networks, which produce an output for a given input, Hopfield networks can recover previously stored patterns from incomplete or noisy inputs by iteratively refining their internal state.

2.5.1 Classical Hopfield Networks

A classical Hopfield network consists of a collection of interconnected neurons whose connections encode a set of stored patterns. During retrieval, the network gradually updates its internal state until it converges to the stored pattern that most closely resembles the input. In this way, the network functions as an associative memory, where partial or corrupted inputs can trigger the recall of complete stored patterns.

Despite their conceptual importance, classical Hopfield networks have several limitations. Their storage capacity is limited, and retrieval performance deteriorates as more patterns are stored. The original formulation operates on binary representations, making it difficult to represent the complex, high-dimensional data commonly encountered in modern machine learning applications. These limitations motivated the development of modern Hopfield networks, which extend the original framework to continuous representations and substantially larger memory capacities.

2.5.2 The Emergence of Modern Hopfield Networks

The limitations of classical Hopfield networks motivated the development of more expressive associative memory architectures. A major advancement was introduced by Ramsauer et al. with the modern Hopfield network, which replaces discrete attractor states with memories represented in continuous high-dimensional spaces [Ramsauer et al. 2021].

Instead of storing memories implicitly within fixed network parameters, modern Hopfield networks represent memories explicitly as continuous vectors. Retrieval is performed by comparing an input query representation against stored memory representations, allowing the system to identify relevant memories based on similarity.

An important result of this formulation is its connection to the attention mechanism used in Transformer models. Ramsauer et al. showed that the update rule of modern Hopfield networks is mathematically equivalent to the attention mechanism, establishing attention as a form of continuous associative memory retrieval [Ramsauer et al. 2021; Vaswani et al. 2017].

2.5.3 External Memory Representation

A key distinction between classical and modern Hopfield networks is the separation between model parameters and stored memories. Classical Hopfield networks encode memories within the connection weights of the network, meaning that adding new memories requires modifying the model itself.

Modern Hopfield networks instead maintain memories as an external set of representations. This enables memories to be added or updated independently of the underlying retrieval mechanism, avoiding the need to retrain the model when new information becomes available.

This property makes modern Hopfield networks particularly relevant for adaptive machine learning systems, where the stored knowledge may change over time while the core model remains unchanged.

2.5.4 Summary

Classical Hopfield networks introduced the concept of associative memory but were limited by discrete memory representations and restricted storage capacity. Modern Hopfield networks extend this concept by leveraging continuous representation spaces and external memory storage, enabling scalable and dynamically updateable associative memory. These properties provide a foundation for machine learning systems that separate the process of retrieving information from the storage of memories.

2.6 Associative Memory in Biological and Artificial Systems

Associative memory refers to the ability of a system to store and retrieve information based on relationships between patterns. In artificial intelligence (AI), associative memory models are designed to recover stored information from partial or incomplete inputs by learning associations between representations. Early approaches, such as Hopfield Networks, modelled memories as stable states within interconnected networks, where the structure of connections influenced memory storage and retrieval [Hopfield 1982]. Modern Hopfield Networks (MHNs) build upon this foundation by introducing high-dimensional representations and attention-based retrieval mechanisms, allowing associative memory models to achieve greater scalability and improved retrieval capabilities [Ramsauer et al. 2021].

The concept of associative memory is also observed in biological neural systems, where it supports processes such as learning, recognition, and recall. In the human brain, memories are represented through distributed patterns of neural activity across interconnected networks rather than being stored in individual neurons [Kan13]. The connections between neurons are formed through synapses, whose strengths can change through synaptic plasticity, allowing the brain to adapt based on experience. One proposed mechanism for this adaptation is Hebbian learning, where repeated co-activation between neurons strengthens their connections and contributes to the formation of associations between patterns [Heb49].

2.7 Bi-Encoder Models

Bi-encoder architectures are neural representation models designed to encode two inputs independently into continuous vector representations. In the context of information retrieval and compliance-oriented document search, bi-encoders provide an efficient mechanism for mapping queries and documents into a shared semantic embedding space. Rather than relying on manually defined matching rules or lexical overlap, bi-encoders learn a retrieval function that determines the relevance between queries and documents through learned representations. This enables large-scale retrieval by comparing vector representations instead of performing expensive pairwise interactions between every query and document [Kar20].

The emergence of transformer-based language models has significantly improved the quality of learned retrieval functions by enabling dense representations that capture contextual meaning. Unlike traditional sparse retrieval approaches, which rely primarily on exact term matching, transformer-based encoders exploit self-attention mechanisms to model relationships between words and concepts within their context [Vaswani et al. 2017]. As a result, semantically related concepts can be represented closely within the embedding space, even when their lexical forms differ [RG19].

2.7.1 Architecture of a Bi-Encoder

A bi-encoder learns separate encoding functions that transform a query q and a document d independently into dense representations:

$$h_q = E_q(q) \tag{2.7}$$

$$h_d = E_d(d) \tag{2.8}$$

where E_q and E_d denote the query and document encoders, respectively, and h_q and h_d represent their resulting embeddings.

The learned retrieval function estimates the relevance between a query and document by measuring the similarity between their corresponding representations. A common similarity measure is cosine similarity:

$$sim(q, d) = \frac{h_q \cdot h_d}{||h_q||||h_d||} \tag{2.9}$$

Through training, the model learns an embedding space where relevant documents are positioned closer to their associated queries, while irrelevant documents are separated. Therefore, retrieval can be interpreted as a learned nearest-neighbor search problem over stored representations rather than a manually specified matching process.

The independent encoding process allows document embeddings to be computed in advance and stored in an index. During retrieval, only the query representation must be generated, after which efficient nearest-neighbor search can identify relevant documents [Kar20]. This property makes bi-encoders particularly suitable for large-scale retrieval scenarios where exhaustive comparison between queries and documents is computationally impractical.

2.7.2 Bi-Encoders in Two-Stage Retrieval Systems

A major advantage of bi-encoder architectures is their computational efficiency. However, the independent encoding process limits their ability to model detailed interactions between a query and document. Since the query and document representations are created separately, fine-grained relationships between specific terms, entities, or contextual dependencies may not be fully captured [NC19].

To address this limitation, modern retrieval systems commonly employ a two-stage architecture. The first stage uses a bi-encoder as a candidate generation mechanism, retrieving a manageable subset of potentially relevant documents from a large corpus. These candidates are subsequently evaluated by a more computationally expensive interaction model, such as a cross-encoder, which performs deeper query-document reasoning [NC19].

This design creates a balance between scalability and accuracy. In compliance applications, this property is particularly relevant because regulatory systems may need to search across large collections of regulations, policies, contracts, and internal documentation while maintaining sufficient retrieval quality.

2.7.3 Relationship to Attention-Based Memory Models

The learned embedding space produced by bi-encoders can be interpreted as a structured semantic memory, where documents are organized according to their learned relationships with possible queries. Each document representation acts as a stored pattern within this space, and retrieval occurs by identifying representations that are most relevant to an input query.

This perspective creates a connection between dense retrieval models and attention-based memory architectures, such as modern Hopfield networks. Instead of presenting an entire document collection to computationally expensive downstream models, bi-encoders can function as an initial retrieval mechanism that selects a smaller set of relevant candidates. This reduces the number of samples requiring further evaluation and allows more complex attention-based models to focus their computation on the most informative representations.

Modern Hopfield networks extend classical associative memory by using attention-based retrieval mechanisms to identify relevant stored patterns [Ramsauer et al. 2021]. Similar to transformer attention, these networks retrieve information by measuring relationships between a query representation and stored memory patterns. The learned retrieval function of a bi-encoder provides a compatible representation space in which associative memory mechanisms can efficiently operate.

For compliance-oriented applications, this approach is especially valuable due to the large volume and complexity of regulatory information. By using bi-encoders for candidate generation, only the most semantically relevant documents need to be evaluated by subsequent models, reducing computational requirements while maintaining retrieval quality. This enables attention-based models and human reviewers to operate on a focused set of high-value candidates rather than processing the entire knowledge base.

2.8 Cross-Encoder Models

Cross-encoder architectures are neural ranking models that jointly process a query and a document to directly estimate their semantic relationship. Unlike bi-encoders, which independently encode inputs into separate vector representations, cross-encoders allow direct interaction between query and document tokens through transformer self-attention mechanisms. This enables more precise relevance estimation by allowing the model to capture fine-grained relationships between input sequences, at the cost of increased computational complexity [NC19; NC20].

In modern retrieval systems, cross-encoders are typically used as a second-stage reranking model. After a bi-encoder retrieves a set of candidate documents, the cross-encoder evaluates these candidates individually and produces a more accurate ranking based on deeper semantic interactions between the query and retrieved passages [NC19; Kar20].

2.8.1 Transformer-Based Interaction Modeling

The primary distinction between bi-encoders and cross-encoders is the point at which interaction occurs. A cross-encoder combines the query and document into a single input sequence:

$$X = q \oplus \delta \oplus d \tag{2.10}$$

where q represents the query, d represents the document, δ represents a delimiter token separating the two inputs, and $\oplus$ denotes sequence concatenation.

The combined sequence is processed by a transformer encoder, allowing every token in the query to attend directly to every token in the document. Through self-attention, the model learns relationships between specific concepts, entities, and contextual information contained in both inputs [Vaswani et al. 2017; Dev19].

The resulting contextual representation can then be passed through a classification or regression layer to produce a relevance score:

$$score(q, d) = f_\theta(q, d) \tag{2.11}$$

where f_θ represents the learned cross-encoder function. This formulation allows the model to directly estimate the relevance of a query-document pair rather than relying on similarity between independently generated embeddings [NC19].

2.8.2 Advantages and Limitations

The primary advantage of cross-encoders is their ability to capture fine-grained semantic interactions. Because the query and document are processed together, the model can identify relationships that independent embeddings may overlook. This makes cross-encoders particularly effective in domains where subtle contextual distinctions determine relevance [NC19].

For compliance-related applications, this capability is important because regulatory interpretation often depends on contextual relationships. A relevant document may not contain identical terminology to the query but may describe equivalent obligations, restrictions, or procedural requirements. Transformer-based interaction models are well suited for these tasks because they can capture contextual meaning beyond direct lexical overlap [Dev19].

However, the increased interaction between inputs also introduces significant computational costs. Since every query-document pair must be processed independently, applying a cross-encoder directly to an entire document collection is computationally infeasible for large-scale retrieval tasks [NC19].

Therefore, cross-encoders are generally applied after an initial retrieval stage performed by a more efficient model such as a bi-encoder. This two-stage architecture enables efficient candidate retrieval while preserving the ranking accuracy of interaction-based models [Kar20].

2.8.3 Cross-Encoders and Attention-Based Retrieval

Cross-encoders demonstrate the importance of attention-based interaction in understanding semantic relationships. The self-attention mechanism allows the model to dynamically determine which parts of a document are relevant to specific parts of a query. This ability emerges from the attention mechanism, where token representations are updated based on their relationships with other tokens in the sequence [Vaswani et al. 2017].

This interaction-based retrieval paradigm shares conceptual similarities with modern Hopfield networks. Both approaches rely on attention-driven retrieval mechanisms to identify relevant patterns from stored representations. Modern Hopfield networks extend classical associative memory by introducing continuous states and attention-based retrieval dynamics, providing a theoretical connection between attention mechanisms and memory retrieval [Ramsauer et al. 2021].

In modern Hopfield networks, attention determines which stored memories are retrieved based on similarity between query patterns and stored representations. Similarly, a cross-encoder determines document relevance by modeling interactions between query and document representations. This connection highlights a broader shift from static feature matching toward dynamic retrieval mechanisms based on learned associations [Ramsauer et al. 2021].

2.8.4 Role in Compliance-Oriented Retrieval Systems

Compliance systems require accurate retrieval because incorrect information retrieval can lead to incomplete assessments or incorrect interpretations of regulatory requirements. While bi-encoders provide efficient candidate generation, cross-encoders improve reliability by performing deeper semantic evaluation of retrieved candidates.

A two-stage retrieval architecture therefore combines the strengths of both approaches:

1. The bi-encoder efficiently retrieves potentially relevant documents from a large knowledge base.

2. The cross-encoder performs detailed semantic analysis and reranks these candidates according to relevance.

This combination provides a practical foundation for compliance applications by achieving both scalability and semantic precision. Furthermore, the attention-based nature of cross-encoders aligns with modern memory-based architectures, including modern Hopfield networks, where retrieval is achieved through learned associations rather than explicit symbolic matching [Ramsauer et al. 2021].

Beyond ranking, the relevance scores produced by cross-encoders can provide an additional mechanism for supporting human oversight in compliance workflows. Since compliance applications often require high levels of accuracy and accountability, automated retrieval systems should be capable of identifying cases where the available evidence is insufficient or where additional expert review is required. The confidence estimates derived from cross-encoder scores can therefore be used to establish review thresholds, allowing high-confidence results to proceed automatically while lower-confidence cases are escalated for human evaluation.

This human-in-the-loop approach is particularly important in regulatory domains, where incorrect retrieval or interpretation may have significant consequences. Rather than replacing expert judgment, cross-encoder-based retrieval systems can support decision-making by reducing the amount of information requiring manual inspection while ensuring that uncertain cases receive additional attention.

2.9 Knowledge Types in Regulatory Compliance

Knowledge can be broadly classified into three categories: explicit, implicit, and tacit knowledge.

2.9.1 Explicit Knowledge

Explicit knowledge refers to information that can be formally documented, stored, and transferred. In the compliance domain, this includes regulations, regulatory guidance, policies, procedures, and control frameworks. Such knowledge represents the codified requirements that AI systems can retrieve, process, and analyse.

2.9.2 Implicit Knowledge

Implicit knowledge refers to knowledge that can be inferred from relationships and patterns within explicit information. In regulatory compliance, obligations often span multiple documents, where relationships between regulations, amendments, supervisory guidance, and internal controls reveal additional meaning that is not directly stated within any single source.

2.9.3 Tacit Knowledge

Tacit knowledge represents experiential knowledge that is difficult to formalise or encode. Within regulatory compliance, it reflects the expertise required to understand how regulatory artefacts relate to one another, including knowing which related documents should be considered, when they should be reviewed, and how changes in one document may affect the interpretation of others.

The regulatory landscape can therefore be represented as a dynamic knowledge graph, where regulations, amendments, technical standards, and guidance documents form interconnected nodes linked through relationships such as "amends", "references", and "clarifies". While explicit knowledge represents the information contained within these nodes, tacit knowledge represents the ability to navigate the relationships between them. For AI systems, capturing this associative knowledge is essential for moving beyond document retrieval towards expert-level regulatory reasoning.

Chapter 3: Research and Artifact Design

3.1 Methodology: Design Science Research

This research follows the *Design Science Research* (DSR) methodology, which provides a systematic approach for designing, developing, and evaluating artifacts intended to address practical problems while contributing to scientific knowledge. Unlike explanatory research approaches that primarily aim to understand existing phenomena, DSR focuses on the creation and evaluation of purposeful artifacts through an iterative design process.

DSR is appropriate for this thesis because the objective is to develop and evaluate an AI-based regulatory entity linking framework. Regulatory documents contain numerous references to legal acts, directives, standards, and technical specifications, where the same regulatory entity may be expressed through different identifiers, abbreviations, titles, or descriptive references. Resolving these references to their corresponding canonical entities is a fundamental requirement for constructing regulatory knowledge graphs and enabling automated compliance workflows.

The research follows the Design Science Research Methodology (DSRM) proposed by Peffers et al. [Pef07], which consists of six iterative activities:

1. **Problem Identification and Motivation.** The research problem is identified through a review of regulatory compliance, entity linking, information retrieval, and natural language processing literature. This analysis reveals limitations in existing approaches for resolving regulatory references due to the domain-specific characteristics of legal documents, including heterogeneous reference formats, semantic ambiguity, cross-references between legal acts, and limited availability of labeled regulatory data.

2. **Definition of Objectives.** Based on the identified problem, the objectives of the artifact are established. The primary objective is to design a framework capable of linking textual regulatory references to their corresponding canonical regulatory entities with high reliability. The framework should support semi-structured regulatory data, incorporate strategies for addressing limited annotations, and enable robust entity resolution across varying reference forms.

3. **Design and Development.** The artifact developed in this research is an end-to-end regulatory entity linking framework covering the stages from regulatory data preparation to model inference. The framework incorporates data preprocessing, data augmentation, semantic representation learning, candidate generation, entity resolution, and the creation of an inference pipeline for applying the developed models to unseen regulatory references. Transformer-based models are utilized to create contextual representations of regulatory references and candidate entities, while multi-stage retrieval approaches are investigated to balance retrieval efficiency and linking accuracy. Additionally, the relationship between Transformer architectures and associative memory models, particularly Modern Hopfield Networks, is examined as a theoretical perspective on the associative retrieval behaviour of these models.

4. **Demonstration.** The developed framework is demonstrated on regulatory datasets containing references to legal entities. Demonstration scenarios focus on resolving different surface forms of regulatory references, including official identifiers, abbreviated names, and descriptive mentions, to their corresponding canonical regulatory entities.

5. **Evaluation.** The developed artifact is evaluated using entity linking metrics and retrieval-based evaluation methods. The evaluation examines the framework's ability to correctly identify regulatory entities, resolve ambiguous references, and generalize across different regulatory reference patterns. The results are analyzed to determine the effectiveness of the proposed framework in supporting reliable automated compliance workflows.

6. **Communication.** The artifact design, implementation process, evaluation results, and research findings are documented throughout this thesis. The conclusions summarize the contributions of the developed regulatory entity linking framework, discuss limitations, and identify opportunities for future research in AI-supported regulatory knowledge management.

The iterative nature of DSR allows insights obtained during implementation and evaluation to guide refinements of the developed artifact. By combining the design of a practical regulatory entity linking framework with systematic evaluation, this research contributes both an artifact for resolving regulatory references and knowledge regarding the application of AI-based entity linking methods in compliance-oriented domains.

3.2 Artifact Requirements

The proposed artifact is a regulatory entity linking framework designed to support the identification and resolution of regulatory references within complex regulatory documents. The following requirements define the expected capabilities of the artifact.

- **Regulatory Reference Resolution:** The artifact shall be capable of linking regulatory references to their corresponding regulatory entities.

- **Semantic Retrieval Capability:** The artifact shall support semantic retrieval approaches capable of identifying relevant regulatory entities beyond exact lexical matching.

- **Scalable Candidate Identification:** The artifact shall efficiently reduce the search space of possible regulatory entities to enable the application of more computationally expensive semantic models.

- **Accurate Entity Ranking:** The artifact shall provide a mechanism for evaluating and ranking candidate regulatory entities based on their semantic relevance.

- **Robustness to Regulatory Variations:** The artifact shall account for variations in regulatory references and formatting.

- **Reliability Assessment:** The artifact shall provide confidence estimates to support reliable deployment in compliance-oriented scenarios and enable human verification of uncertain predictions.

- **Evaluation Capability:** The artifact shall support quantitative evaluation of retrieval performance and final regulatory entity linking accuracy.

3.3 Proposed Framework

This section introduces the proposed framework for regulatory entity linking. The framework integrates transformer-based dense retrieval, cross-encoder reranking, and large language model (LLM)-based data augmentation to improve the identification and resolution of regulatory references within complex regulatory documents.

The proposed framework consists of four primary components:

1. Data preparation and construction of a regulatory reference dataset

2. Development of dense retrieval representations using a bi-encoder architecture

3. Domain adaptation through hard negative mining and noise-aware data augmentation

4. Two-stage regulatory entity linking through candidate retrieval and semantic reranking

An overview of the final inference architecture is presented in Figure 3.1. The following sections describe the individual components of the proposed framework.

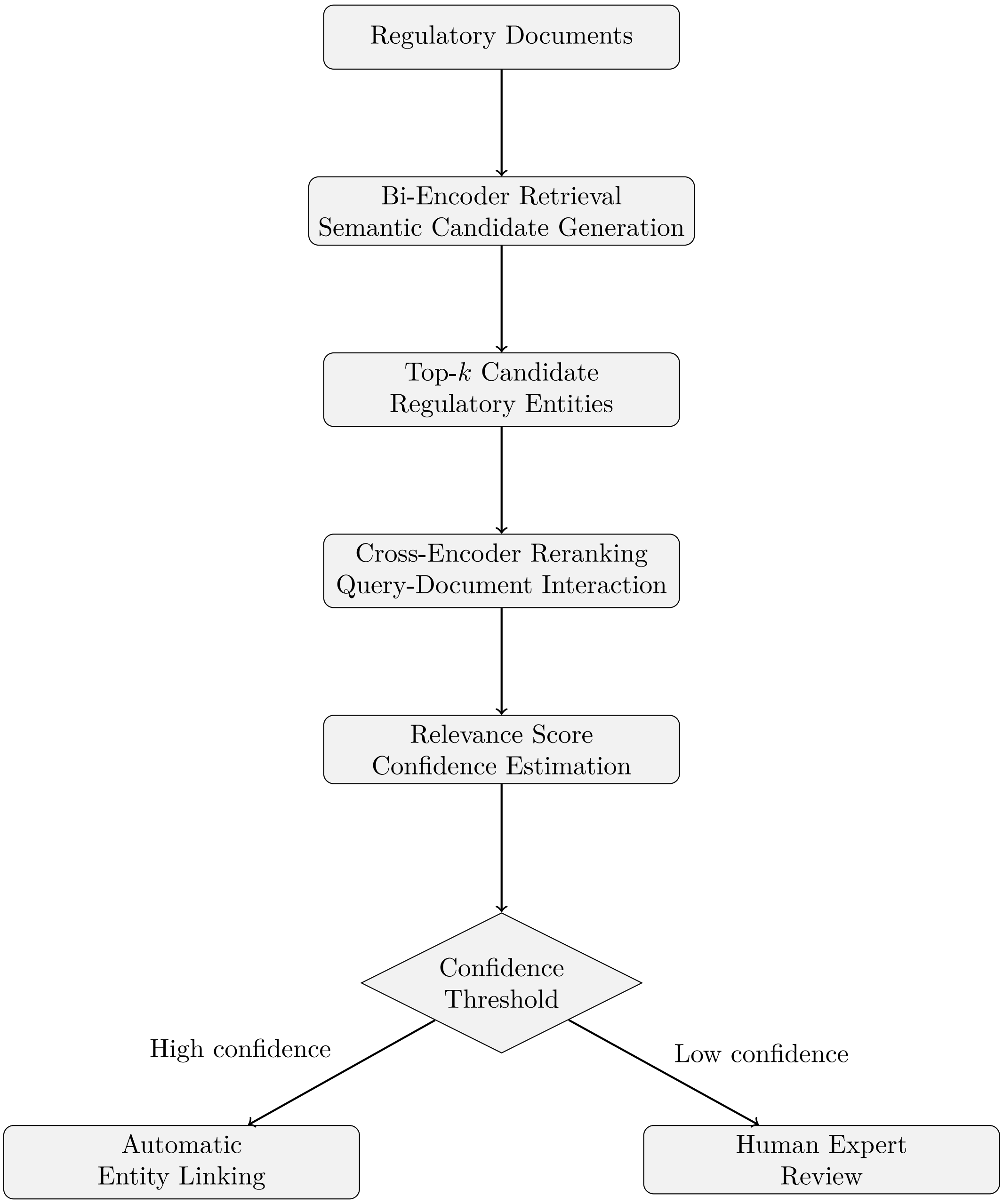

Figure 3.1: Conceptual overview of the proposed regulatory entity linking framework.

3.3.1 Data Preparation

The proposed framework begins with a data preparation stage responsible for transforming raw regulatory documents into structured representations suitable for machine learning models. This stage establishes the necessary inputs for subsequent retrieval and reranking components.

The framework incorporates the extraction of regulatory references, contextual information, and relevant document characteristics to support the identification of relationships between references and regulatory entities. These representations provide the foundation for generating training examples and enabling transformer-based models to learn meaningful semantic relationships.

3.3.2 Construction of the Regulatory Reference Dataset

A central component of the proposed framework is the construction of a regulatory reference dataset that represents associations between regulatory references and their corresponding entities.

Due to the structured characteristics of regulatory references, the framework incorporates heuristic-based approaches to identify reference patterns and establish initial relationships between mentions and regulatory entities. These relationships provide the supervision required for training retrieval models while reducing the dependency on extensive manual annotation.

The framework further incorporates a dataset partitioning strategy to enable reliable model evaluation. The dataset is separated into training, validation, and test subsets to support model optimization, configuration selection, and final performance assessment.

To evaluate generalization under realistic compliance scenarios, the framework considers document-level separation between dataset partitions. This ensures that evaluation is performed on regulatory documents that are not observed during model development, allowing the framework to measure its ability to resolve references within previously unseen documents.

3.3.3 Embedding Model Development

The first retrieval component of the proposed framework is a bi-encoder model designed to learn dense semantic representations of regulatory references and candidate entities.

The bi-encoder functions as an efficient candidate generation mechanism by mapping references and entities into a shared embedding space. This enables large-scale retrieval through similarity-based comparison rather than exhaustive evaluation of all possible reference-entity combinations.

To improve retrieval performance, the framework incorporates iterative hard negative mining. During this process, highly similar but incorrect candidates are identified and incorporated into subsequent training iterations. These challenging examples encourage the embedding model to learn more discriminative representations and improve its ability to distinguish between closely related regulatory entities.

The resulting bi-encoder provides the candidate retrieval stage of the proposed two-stage architecture.

3.3.4 Domain Noise Characterization

Regulatory documents frequently contain variations that may not be represented within an initial training dataset. These variations can originate from differences in formatting, terminology, citation styles, and document-specific writing conventions.

To address this challenge, the proposed framework incorporates an LLM-based domain noise characterization stage. This stage is designed to analyze unlabeled regulatory documents and identify recurring sources of variation within regulatory references.

The identified variations are represented through a noise taxonomy describing common deviations observed in regulatory data. Examples of such variations include:

- abbreviated regulatory references
- incomplete citations
- typographical variations
- formatting inconsistencies
- organization-specific terminology

The resulting taxonomy provides an abstract representation of domain-specific noise patterns. By capturing general characteristics of regulatory variation rather than individual examples, the taxonomy enables the framework to incorporate realistic data variability without directly transferring specific regulatory references or document-specific information from the unlabeled corpus into the training data.

3.3.5 LLM-Based Data Augmentation

The proposed framework incorporates LLM-based data augmentation to introduce realistic variations into the constructed training dataset. The generated noise taxonomy is used to guide the augmentation process by defining the types of variations that should be simulated.

The LLM applies the identified noise patterns to existing regulatory reference examples while preserving their underlying semantic relationships. This enables the generation of additional training instances that represent potential variations encountered within real-world regulatory documents.

By separating noise characteristics from the individual regulatory references in which they occur, the augmentation process is designed to avoid directly incorporating information from the unlabeled corpus. Instead, the generated examples reflect general patterns of regulatory variation, allowing the framework to increase training diversity while maintaining separation between the original data sources and the generated training instances.

Through noise-aware augmentation, the framework aims to provide retrieval models with exposure to a broader range of regulatory reference expressions and improve their ability to handle previously unseen variations.

3.3.6 Cross-Encoder Development

The final retrieval stage incorporates a cross-encoder model for fine-grained evaluation of candidate regulatory entities.

Following candidate generation by the bi-encoder, the cross-encoder jointly processes the regulatory reference and candidate entity to model detailed semantic interactions. This enables more accurate relevance estimation by considering relationships that may not be captured through independent embeddings.

The framework further incorporates hard negative mining for cross-encoder optimization. Challenging candidates retrieved by the bi-encoder are used to improve the model's ability to differentiate between highly similar regulatory entities.

The combination of efficient candidate generation and interaction-based reranking provides a scalable approach for regulatory entity linking while maintaining high semantic precision.

3.3.7 Confidence-Based Human Oversight

Due to the high reliability requirements of compliance applications, the proposed framework incorporates a confidence-based decision mechanism. Predictions exceeding a predefined confidence threshold can be automatically resolved, while uncertain predictions can be forwarded for human verification.

Although human review is primarily intended as a reliability mechanism, the collected feedback can provide a valuable source of additional training data. In particular, difficult or ambiguous regulatory references identified during review may be incorporated into future training iterations to improve model robustness.

3.4 Design Decisions

The proposed regulatory entity linking framework is guided by a series of architectural design decisions intended to address the challenges of semantic retrieval within compliance-oriented environments. The following decisions establish the conceptual foundation of the proposed artifact and motivate the selection of its principal components.

3.4.1 Adoption of a Two-Stage Retrieval Architecture

Regulatory corpora may contain thousands of potential candidate entities, making exhaustive interaction between every regulatory reference and every candidate computationally infeasible. Consequently, the proposed framework adopts a two-stage retrieval architecture consisting of an initial retrieval stage followed by a semantic reranking stage.

The first stage is responsible for efficiently retrieving a small set of semantically relevant candidates, while the second stage performs detailed interaction-based relevance estimation. This separation balances computational efficiency with retrieval accuracy and enables scalable deployment on large regulatory collections.

3.4.2 Dense Semantic Retrieval

Regulatory references frequently exhibit considerable lexical variation resulting from abbreviations, formatting differences, incomplete citations, and organization-specific terminology. Traditional lexical retrieval methods rely primarily on exact token overlap and therefore struggle when equivalent regulatory concepts are expressed differently.

For this reason, the proposed framework adopts dense semantic representations learned by transformer-based language models. Mapping regulatory references and entities into a shared embedding space enables retrieval based on semantic similarity rather than lexical correspondence, improving robustness to variations commonly observed within regulatory documents.

3.4.3 Iterative Hard Negative Mining

Many incorrect regulatory entities are easily distinguishable from the correct entity and therefore provide limited training value. Instead, the proposed framework incorporates iterative hard negative mining to identify candidates that are semantically similar yet incorrect.

Training on these difficult examples encourages the retrieval models to learn more discriminative representations and improves their ability to distinguish between highly related regulatory entities. Hard negative mining is therefore incorporated as an iterative refinement strategy throughout model development.

3.4.4 Noise-Aware Data Augmentation

The distribution of regulatory references encountered during deployment is unlikely to be fully represented by the initial training dataset. Rather than applying generic augmentation techniques, the proposed framework characterizes recurring sources of variation within unlabeled regulatory documents and represents them through an abstract taxonomy of domain-specific noise.

This taxonomy describes common patterns of variation without directly incorporating individual regulatory references from the unlabeled corpus. Consequently, it enables realistic augmentation while reducing the risk of introducing document-specific information into the generated training data. The resulting synthetic examples are intended to improve robustness against the variability encountered in practical compliance environments.

3.4.5 Confidence-Based Decision Support

Regulatory compliance applications often require a high degree of reliability because incorrect entity linking may influence downstream compliance assessments. For this reason, the proposed framework incorporates confidence estimation as part of the final decision process.

Rather than treating all predictions equally, the framework is designed to distinguish between high-confidence and uncertain predictions. High-confidence predictions may be accepted automatically, whereas uncertain predictions can be forwarded for human verification. This design supports practical deployment by combining automated retrieval with human oversight where additional assurance is required.

Chapter 4: Implementation

This chapter describes the implementation of the proposed regulatory entity linking framework. While the previous chapter introduced the conceptual design of the artifact, this chapter details the practical realization of the individual components, including data preparation, retrieval model development, augmentation strategies, and the experimental infrastructure used for evaluation. The implementation follows a two-stage retrieval architecture consisting of a dense retrieval component and a semantic reranking component. The first stage utilizes a bi-encoder model to efficiently retrieve potentially relevant regulatory entities from a large candidate space. The second stage applies a cross-encoder model to perform fine-grained relevance estimation between retrieved candidates and regulatory references.

4.1 System Architecture

The implemented system follows the two-stage retrieval architecture described in the proposed framework. The objective is to efficiently identify regulatory entities corresponding to extracted regulatory references while maintaining sufficient semantic accuracy for compliance-oriented applications.

The implemented architecture consists of four primary components:

1. Regulatory reference processing and representation
2. Dense candidate retrieval using a bi-encoder
3. Semantic candidate reranking using a cross-encoder
4. Confidence-based decision support

The bi-encoder provides an efficient retrieval mechanism by mapping regulatory references and candidate entities into a shared embedding space. Retrieved candidates are subsequently evaluated by the cross-encoder, which performs deeper interaction-based relevance estimation.

4.1.1 Model Components

The retrieval component is implemented using a transformer-based bi-encoder architecture. The model receives regulatory references and candidate entities as independent inputs and generates dense vector representations used for similarity-based retrieval.

The reranking component is implemented using a transformer-based cross-encoder architecture. Unlike the bi-encoder, the cross-encoder jointly processes the regulatory reference and candidate entity, allowing direct token-level interaction through self-attention.

4.1.2 Experimental Environment

The experiments were conducted using the implemented training and evaluation pipeline. The system was developed using the PyTorch deep learning framework together with transformer-based model implementations. The computational environment used throughout model development and evaluation is summarized in Table 4.1.

The computational environment was kept consistent across experiments to ensure comparable evaluation results.

Table 4.1: Experimental environment used for model development and evaluation.

Component	**Specification**
GPU	NVIDIA RTX 4070 (12GB VRAM)
CPU	AMD Ryzen 7 5800X
Memory	32GB DDR4 RAM
Operating System	CachyOS Linux
Frameworks	PyTorch, Hugging Face Transformers

The computational environment was kept consistent across experiments to ensure comparable evaluation results.

4.1.3 Model Selection and Configuration

The proposed framework consists of multiple transformer-based components, each serving a distinct role within the regulatory entity linking pipeline. Table 4.2 summarizes the pretrained models used as initialization points and their respective purposes within the framework.

Table 4.2: Models used within the implemented regulatory entity linking framework.

Component	Base Model
Embedding Model	google/embeddinggemma-300m [Ver25]
Cross-Encoder	BAAI/bge-reranker-v2-m3 [Che24]
Large Language Model	Gemma-4-12B-it [Gem26] Unsloth UD-Q4_K_XL GGUF Quantization [Uns26]

Figure 3.1 provides an overview of the implemented system architecture and illustrates the interaction between the individual components of the regulatory entity linking pipeline.

The bi-encoder serves as the initial candidate retrieval component and is fine-tuned on regulatory reference-entity pairs to learn domain-specific semantic representations. The reranking model is subsequently fine-tuned to perform fine-grained relevance estimation between retrieved candidates and regulatory references. Additionally, a large language model is utilized for domain adaptation through noise characterization and data augmentation.

The system receives regulatory documents as input and extracts relevant regulatory references. These references are encoded into dense representations and compared against indexed regulatory entities to retrieve a set of candidate matches. The candidate set is subsequently evaluated by the cross-encoder, which produces relevance scores used for final entity linking decisions.

4.1.4 Data Preparation and Feature Selection

The dataset used in this work was provided by Fairfield & Archer, a partner organization specializing in regulatory compliance solutions. The dataset consists of structured regulatory information extracted from compliance documents and was provided for research and evaluation purposes.

The provided dataset serves as the initial input for the regulatory entity linking framework.

4.1.5 Feature Selection

For the regulatory entity linking task, only features relevant to identifying relationships between regulatory references and regulatory entities are retained.

The selected features represent the information required to identify relationships between regulatory references and their corresponding entities. The main features used within the framework are described below.

Table 4.3 summarizes the selected features and their role within the linking process.

Table 4.3: Selected features used for regulatory entity linking.

Pair Item 1: Regulatory Reference

Feature	**Description**
Source Build ID	Identifier of the source document or dataset entry.
Reference Title	The title of the document being refered to, as defined in the reference
Description	Description, that describes how the source document is related to the document being referred to.
Relationship Type	The type of relationship the source document has to the document being referred to.

Pair Item 2: Regulatory Entity

Feature	**Description**
Build ID	Identifier of the Regulatory Document Entity
Short Title	shortened title of the Regulatory Document Entity
Aliases	Alternative names associated with the entity.
Document Type	The type of document being described

4.2 Regulatory Reference Dataset Construction

Following data preparation, a regulatory reference dataset is constructed to provide the training and evaluation examples required by the retrieval models. The objective of this process is to establish associations between extracted regulatory references and their corresponding regulatory document entities.

4.2.1 Heuristic Data Labelling

Due to the scale of the regulatory corpus, manual annotation of reference-entity associations is impractical. Therefore, the dataset construction process relies on a set of heuristic matching strategies to automatically establish candidate associations. These heuristics utilize available regulatory entity attributes, including document titles, short titles, aliases, and extracted regulatory identifiers.

The matching process consists of multiple complementary strategies designed to capture different forms of regulatory references. Exact matching strategies identify direct correspondences between extracted reference titles and regulatory entity attributes, while more flexible strategies account for abbreviated references, aliases, and references containing document identifiers.

The overall heuristic matching process is summarized in Algorithm 1.

Algorithm 1 Heuristic regulatory reference matching

Require: Regulatory references R, regulatory entities D

for each reference $r \in R$ **do**
    Generate candidate matches using title, short title, and alias matching
    Extract regulatory identifiers from r
    Compare extracted identifiers with entity identifiers
    Rank candidate matches using contextual matching rules
    Store highest-scoring valid match
end for
return Reference-entity associations

The identifier-based matching strategy provides additional coverage for references where direct textual similarity is insufficient. Regulatory identifiers are extracted from both reference mentions and regulatory entity titles, and matches are determined based on identifier overlap and contextual relationships. For example, references representing the identity of a regulatory document require exact identifier agreement, whereas references describing relationships such as amendments, repeals, or implementing acts are matched through shared regulatory identifiers and contextual constraints.

The generated associations are categorized according to the heuristic strategy responsible for identifying the match. Table 4.4 summarizes the distribution of examples produced by each matching strategy.

Table 4.4: Distribution of heuristic matching strategies used for regulatory reference linking.

Matching Strategy	**Number of Examples**
Exact title matching	X
Short title matching	X
Alias matching	X
Identifier-based matching	X
Other pattern-based matching	X

The resulting reference-entity associations are used to construct training examples for the retrieval models. Positive examples consist of references paired with their corresponding regulatory entities. Negative examples are generated during subsequent retrieval model training through candidate sampling and hard negative mining procedures.

4.2.2 Dataset Partitioning

To evaluate the ability of the proposed framework to generalize to previously unseen regulatory entities, the constructed dataset is partitioned into training, validation, and test subsets using entity-level separation.

Rather than randomly splitting individual reference examples, all examples associated with the same regulatory document entity are assigned to a single partition. This prevents references associated with identical regulatory entities from appearing across multiple subsets.

This separation strategy reduces information leakage between training and evaluation data and ensures that evaluation measures the ability of the model to resolve references associated with regulatory entities that were not observed during training.

The resulting dataset distributions after splitting are summarized in Table 4.5.

Table 4.5: Distribution of regulatory reference examples across dataset partitions.

Dataset Split	Number of Examples
Training	X
Validation	X
Test	X

4.2.3 Pair Transformation

Following dataset construction and partitioning, the reference-entity associations are transformed into textual representations suitable for training the retrieval models. The transformation process converts structured regulatory references and regulatory entities into standardized text formats while preserving the information relevant for semantic matching.

Each training example consists of a reference document representation and a target regulatory entity representation. The reference representation contains the extracted reference title, relationship type, and descriptive information associated with the reference. The target representation contains the regulatory entity title, metadata, and known aliases.

The resulting pair structure enables the retrieval models to learn semantic relationships between regulatory references and their corresponding entities. An example transformed training pair is shown in Figure 4.2.

```
[REFERENCE DOCUMENT]
Title:  Commission Delegated Regulation (EU) 2025/753

[LINK TYPE]
amendment

[LINK DESCRIPTION]
The source Decision amends Annex IX (Financial services) to the EEA
Agreement by incorporating Commission Delegated Regulation (EU) 2025/753,
making its provisions applicable within the EEA. The Decision also
specifies the authentic language versions for Icelandic and Norwegian.

--------------------------------------------------------------------------

[TARGET DOCUMENT]
Title:  Commission Delegated Regulation (EU) 2025/753 of 16 April 2025
supplementing Regulation (EU) 2023/2631 of the European Parliament and of
the Council by establishing the content, methodologies, and presentation
of the information to be voluntarily disclosed by issuers of bonds
marketed as environmentally sustainable or of sustainability-linked bonds
in the templates for periodic post-issuance disclosures

[DOCUMENT METADATA]
Short title:  Delegated Regulation (EU) 2025/753
Document type:  regulatory_document

[ALIASES]
- EU Green Bond Voluntary Post-Issuance Disclosure Delegated Regulation
- Commission Delegated Regulation (EU) 2025/753
- EU Sustainable Bond Disclosure Templates Regulation
```

Figure 4.2: Example transformed reference-entity training pair used as input for retrieval model training.

The transformed representations are subsequently used as inputs for model training. For the bi-encoder model, reference and entity representations are encoded independently to learn dense semantic representations. For the cross-encoder model, both representations are jointly processed to estimate the relevance between a reference and candidate entity pair.

4.3 Bi-Encoder Training

Following the dataset creation, the bi-encoder component is fine-tuned on the constructed regulatory reference dataset to learn domain-specific semantic representations for regulatory reference retrieval. The model is fully fine-tuned, allowing the pretrained embedding parameters to be adapted directly to the regulatory entity linking task.

The implementation uses the SentenceTransformers library, which provides training utilities for dense retrieval models and retrieval-oriented evaluation methods [RG19]. The model is optimized using Cached Multiple Negatives Ranking Loss (CachedMNRL). This loss function learns representations from positive reference-entity pairs by treating other examples within the batch as implicit negative examples. During training, the model maximizes the similarity between matching reference-entity pairs while minimizing similarity with non-matching pairs within the batch.

The effectiveness of Multiple Negatives Ranking Loss is strongly influenced by the number of available in-batch negatives. Larger effective batch sizes provide more negative examples, increasing the difficulty of the retrieval task and encouraging the model to learn more discriminative embeddings. To support larger effective batch sizes under limited GPU memory, CachedMNRL stores representations from previous mini-batches, allowing additional negatives to be incorporated without requiring all examples to be processed simultaneously [Hen17].

To ensure meaningful negative examples, a no-duplicate batch sampler is used. This prevents multiple examples belonging to the same regulatory entity from appearing within a single batch, ensuring that in-batch negatives represent distinct candidate entities.

The model is trained using mixed-precision BF16 computation. The maximum sequence length is set to 512 tokens to accommodate longer regulatory references and contextual information while maintaining computational efficiency.

Due to the memory requirements associated with full model fine-tuning, gradient checkpointing is enabled using `gradient_checkpointing_enable()`. Gradient checkpointing reduces memory consumption by avoiding storage of intermediate activations during the forward pass and instead recomputing selected activations during backpropagation. This approach reduces GPU memory requirements at the cost of additional computation time [Che16].

The optimizer configuration uses AdamW with 8-bit optimizer states provided through the bitsandbytes library. The 8-bit implementation maintains comparable optimization behaviour to standard AdamW while reducing memory consumption by representing optimizer states using lower-precision 8-bit data types instead of traditional 32-bit floating-point representations. This reduction in optimizer-state memory enables larger effective batch sizes and more efficient fine-tuning on constrained GPU hardware [Det22].

Gradient accumulation is used to increase the effective batch size without exceeding GPU memory limitations. This is particularly relevant for Multiple Negatives Ranking Loss, where the number of negative examples available during optimization is directly related to the batch size. The effective batch size is determined by the product of the physical batch size and the number of gradient accumulation steps.

The main training configuration is summarized in Table 4.6.

Table 4.6: Bi-encoder training configuration.

Parameter	**Configuration**
Training approach	Full fine-tuning
Precision	BF16 mixed precision
Maximum sequence length	512 tokens
Loss function	Cached Multiple Negatives Ranking Loss
Batch sampling	No-duplicate batch sampler
Optimizer	AdamW 8-bit (bitsandbytes)
Learning rate	2×10^{-5}
Weight decay	0.01
Warmup ratio	0.05
Gradient checkpointing	Enabled
Gradient accumulation steps	4
Evaluation framework	SentenceTransformers InformationRetrievalEvaluator
Validation metric	Cosine MRR@10

During training, model performance is monitored using the SentenceTransformers InformationRetrievalEvaluator on the validation dataset. The primary validation metric is:

```
eval_validation-document-linking_cosine_mrr@10
```

which measures the mean reciprocal rank of the correct regulatory entity among the top-k retrieved candidates using cosine similarity.

The validation metric is used only for monitoring training progress and model selection. Final performance evaluation is conducted separately using an independent test dataset outside of the training loop to provide an unbiased assessment of retrieval performance on unseen regulatory entities.

4.3.1 Hard Negative Mining

Following the initial training phase of the bi-encoder, a hard negative mining procedure was utilized to generate additional training examples for continued fine-tuning. The procedure was introduced to provide more challenging negative examples than those obtained through random sampling.

The trained bi-encoder was used to encode the regulatory entity collection into dense vector representations, which were indexed for similarity-based retrieval. For each regulatory reference, the corresponding embedding was generated and used to retrieve the top-k most similar regulatory entities.

The ground-truth regulatory entity was removed from the retrieved candidate set, and the highest-ranked remaining entities were selected as hard negatives. These candidates represented semantically similar but incorrect regulatory entities, such as documents with overlapping titles, related regulatory terminology, or similar identifiers.

To prevent incorrect negative assignments, retrieved candidates were compared against the labelled reference-entity associations generated during dataset construction. Candidates corresponding to valid reference-entity relationships were excluded before being added to the hard negative set.

The resulting hard negative examples were incorporated into the training dataset and used during continued fine-tuning of the bi-encoder. This expanded training procedure allowed the model to learn improved separation between closely related regulatory entities.

4.4 LLM Augmentation

LLM augmentation was used to generate alternative representations of regulatory references while preserving the original reference-entity association. The objective of this augmentation step is to increase robustness against variations in how regulatory entities are referenced in real-world compliance documents.

The original reference instances were provided as input to a large language model, together with predefined noise categories, which described possible variations in regulatory references. The model was instructed to generate semantically equivalent representations while maintaining the same underlying regulatory entity association.

The augmentation process was designed as a controlled transformation step rather than an open-ended generation process. The LLM was guided using predefined noise categories and was restricted to modifying the provided reference representations. External regulatory information was not introduced during generation, reducing the risk of unsupported information being incorporated into the training dataset.

The generated variations were incorporated into the training dataset as additional positive examples. Augmentation was performed only after the original dataset was already partitioned into training, validation, and test subsets. This ensured that generated variations derived from training examples could not appear in the evaluation partitions, preventing information leakage. The resulting augmented training set allows the retrieval models to learn from different reference styles, including abbreviated names, alternative terminology, and incomplete regulatory identifiers.

4.4.1 Noise Taxonomy Design

A noise taxonomy was created to capture common variations in regulatory references that may negatively impact retrieval performance. During analysis of the regulatory reference dataset, two primary sources of variation were identified: variations in regulatory document titles and variations in descriptive reference information.

Title variations occur when regulatory entities are referenced using shortened names, alternative naming conventions, identifiers, or modified formatting. Description variations occur when contextual information is omitted, summarized, or reformulated while preserving the underlying relationship between the reference and the regulatory entity.

The taxonomy was designed to provide controlled constraints for the LLM augmentation process. Instead of generating arbitrary synthetic examples, the language model is guided to introduce realistic variations that reflect how regulatory documents are commonly referenced in compliance-oriented text.

The identified noise categories are summarized in Table 4.7.

Table 4.7: Noise taxonomy used for LLM-based regulatory reference augmentation.

Category	**Noise Patterns**
Title variation	Title shortened, alias title, partial title, acronym substitution, abbreviation used, identifier used instead of title, informal short name, word order variation, punctuation variation, parenthetical inclusion, common name substitution.
Identifier variation	Article number extraction, specific article reference, legal act type removal.
Description simplification	Boilerplate removed, date removed, descriptive clauses omitted, document type retained while description removed, key phrase extracted.
Context modification	Organization removed, legal basis extraction, relationship type extraction, procedural context removed, operative event removal, conditional reference extraction.
Domain-specific information removal	Scientific basis omitted, technical specification removal, historical context removal, alias mentioned.

4.4.2 LLM Augmentation Example

The augmentation process transforms an existing reference-entity pair into additional training examples by modifying only the reference representation while preserving the original target entity. The following example demonstrates the transformation of an existing training instance.

The original reference was linked to the following regulatory entity:

```
[ORIGINAL REFERENCE DOCUMENT]
Title: Regulation (EU) 2024/1735 (Net-zero Technology Manufacturing
Ecosystem)

Description: The Regulation references Regulation (EU) 2024/1735
regarding the potential for permanent carbon storage in third countries,
subject to international agreements and equivalent conditions for
geological storage.

[TARGET DOCUMENT]
Title: Regulation (EU) 2024/1735 of the European Parliament and of the
Council of 13 June 2024 on establishing a framework of measures for
strengthening Europe's net-zero technology manufacturing ecosystem and
amending Regulation (EU) 2018/1724

[DOCUMENT METADATA]
Short title: Regulation (EU) 2024/1735
Document type: regulatory_document

[ALIASES]
- Net-Zero Industry Act
- NZIA

--------------------------------------------------------------------------

[AUGMENTED REFERENCE DOCUMENT]
Title: NZIA

Description: Framework for strengthening Europe's net-zero technology
manufacturing ecosystem and amending Regulation (EU) 2018/1724.

[TARGET DOCUMENT]
Title: Regulation (EU) 2024/1735 of the European Parliament and of the
Council of 13 June 2024 on establishing a framework of measures for
strengthening Europe's net-zero technology manufacturing ecosystem and
amending Regulation (EU) 2018/1724

[DOCUMENT METADATA]
Short title: Regulation (EU) 2024/1735
Document type: regulatory_document

[ALIASES]
- Net-Zero Industry Act
- NZIA
```

Figure 4.3: Example of LLM-generated augmentation while preserving the original reference-entity association.

The augmented example differs from the original reference only in its representation of the regulatory entity. The target entity remains unchanged, allowing the generated instance to be incorporated as an additional positive training example.

4.5 Candidate Retrieval

4.6 Cross Encoder training

4.7 Model Harness

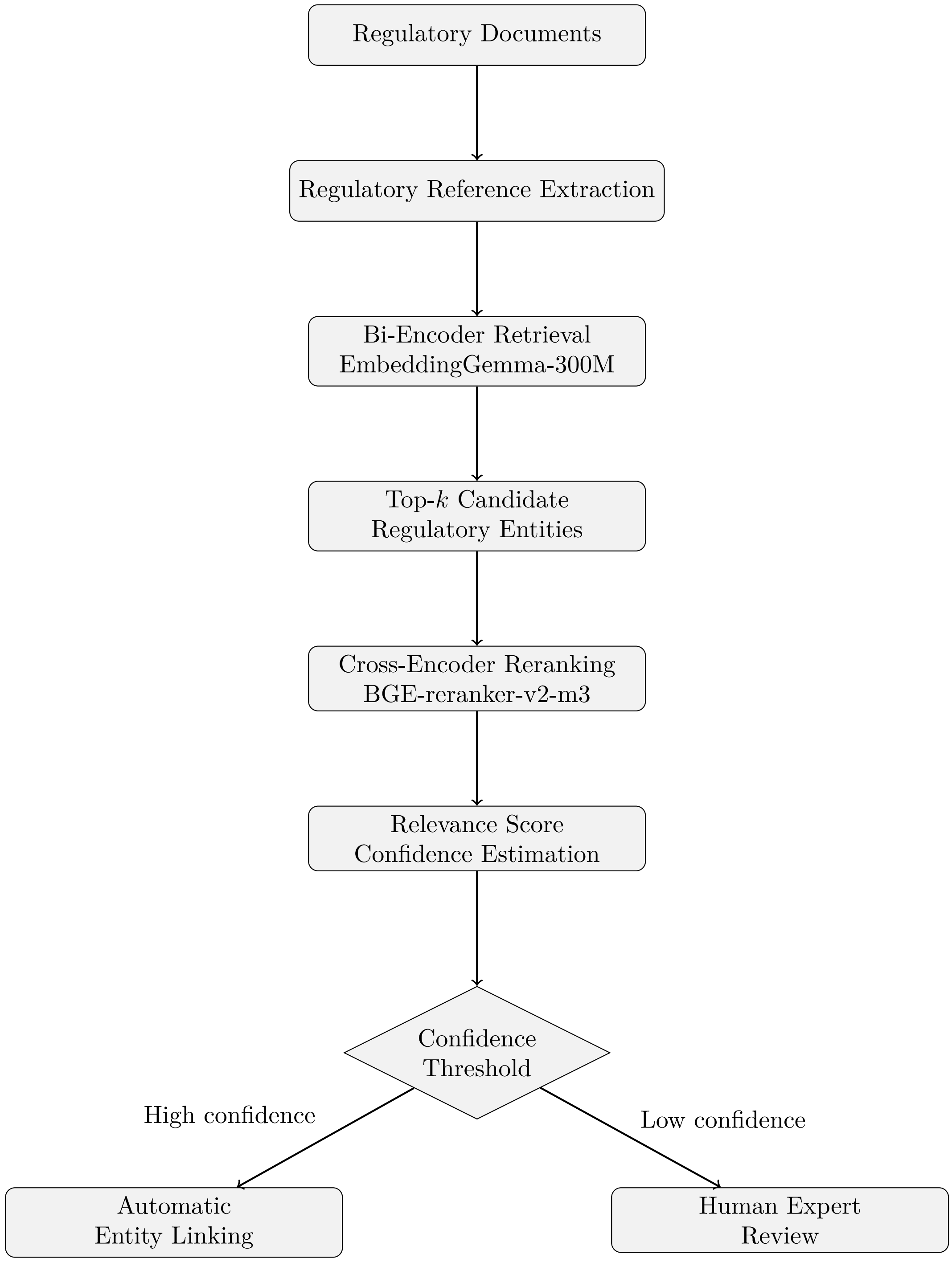

Figure 4.1: Implemented regulatory entity linking architecture.

Chapter 5: Evaluation

5.1 Evaluation Setup and Metrics

The evaluation framework assesses the performance of the implemented regulatory entity linking pipeline. Evaluation is performed using the held-out test partition created during dataset construction, which contains regulatory entities not observed during training.

The evaluation is divided according to the individual components of the retrieval pipeline. The bi-encoder is evaluated based on its ability to retrieve relevant regulatory entities, while the cross-encoder is evaluated based on its ability to rerank retrieved candidates. Additionally, the complete pipeline is evaluated to measure the final entity linking performance.

The following sections describe the metrics used to evaluate each component.

5.1.1 Bi-Encoder Evaluation

The bi-encoder is evaluated as a candidate retrieval model, where its objective is to retrieve a set of relevant regulatory entities for a given regulatory reference. Since the bi-encoder is used as the first retrieval stage of the pipeline, the evaluation focuses on whether the correct regulatory entity is included within the retrieved candidate set rather than only whether it is ranked first.

The primary metric used for evaluating the bi-encoder is Recall@K. Recall@K measures the proportion of references where the correct regulatory entity is retrieved within the top-K candidates. This metric is suitable for the bi-encoder because the retrieved candidates are subsequently passed to the cross-encoder for reranking. Therefore, a successful bi-encoder should maximize the probability that the correct entity is available for the downstream reranking stage.

$$Recall@K = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(r_i \in C_i^K)$$

where N represents the number of evaluated references, r_i represents the correct regulatory entity for reference i, and C_i^K represents the set of top-K retrieved candidates.

In addition to Recall@K, Mean Reciprocal Rank (MRR) is used to evaluate the ranking quality of retrieved candidates. While Recall@K only considers whether the correct entity is present within the candidate set, MRR considers its position within the ranking. This provides additional insight into whether the bi-encoder assigns higher similarity scores to correct entities.

$$MRR = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{rank_i}$$

where $rank_i$ represents the position of the correct regulatory entity within the retrieved candidate ranking.

5.1.2 Cross-Encoder Evaluation

The cross-encoder is evaluated as a reranking model that receives a regulatory reference together with retrieved candidate entities and predicts their relevance. Unlike the bi-encoder, the objective of the cross-encoder is not candidate discovery but accurate ranking of a smaller candidate set.

The primary metric used for cross-encoder evaluation is Mean Reciprocal Rank (MRR). MRR is suitable for evaluating reranking performance because the objective of the cross-encoder is to assign the highest relevance score to the correct regulatory entity. A higher MRR indicates that the correct entity consistently appears closer to the top of the ranked candidate list.

Recall@K is additionally reported after reranking to measure whether the correct entity remains within the final ranked candidate set. Although candidate recall is primarily determined by the bi-encoder stage, evaluating Recall@K after reranking provides insight into whether the cross-encoder incorrectly demotes relevant candidates.

For binary relevance evaluation, precision, recall, and F1-score are additionally used to evaluate the classification behaviour of the cross-encoder. These metrics measure how effectively the model distinguishes relevant reference-entity pairs from incorrect candidate pairs.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

where TP, FP, and FN represent true positive, false positive, and false negative predictions respectively.

5.2 Entity Linking Performance

5.3 Comparative performance evaluation

5.4 Error Analysis

5.5 Discussion

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Eigenständigkeitserklärung

Hiermit versichere ich, dass ich die vorliegende Bachelorarbeit selbstständig und nur unter Verwendung der angegebenen Quellen und Hilfsmittel verfasst habe. Die Arbeit wurde bisher in gleicher oder ähnlicher Form keiner anderen Prüfungsbehörde vorgelegt.

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